

sequentie: stel 2 rotaties + translatie

$$(x_0, y_0, z_0, 1) \xrightarrow{\text{rot 1}} (x_1, y_1, z_1, 1) \xrightarrow{\text{rot 2}} (x_2, y_2, z_2, 1) \xrightarrow{\text{trans.}} (x_3, y_3, z_3, 1)$$

$$\begin{cases} (x_1, y_1, z_1, 1) = (x_0, y_0, z_0, 1) M_z(\phi_1) \\ (x_2, y_2, z_2, 1) = (x_1, y_1, z_1, 1) M_x(\phi_2) \\ (x_3, y_3, z_3, 1) = (x_2, y_2, z_2, 1) T \end{cases}$$
$$\rightarrow (x_3, y_3, z_3, 1) = (x_0, y_0, z_0, 1) \underbrace{M_z(\phi_1) M_x(\phi_2) T}_{4 \times 4 \text{ matrix}}$$